

Ahmed Thabet

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Engineering Manager with 10+ years building decentralized cloud infrastructure, custom operating systems, and developer platforms — from bare-metal OS design to Kubernetes orchestration at scale.

Experience

- **Codescalers: Engineering Manager** (2023 – Present) Managing multiple engineering teams building the Threefold Grid, a decentralized P2P cloud spanning thousands of nodes across 60+ countries. Own technical strategy, architecture, roadmap, delivery, hiring, and mentoring across the full stack — from a custom Linux-based OS to SDKs, Terraform providers, and web dashboards.
- **Codescalers: Software Development Team Lead** (2020 – 2023) Led development of the Jumpscale cloud framework, the 3Bot virtual system administrator, and the Virtual Data Center platform. Drove system architecture, hands-on development, code quality, and on-time delivery.
- **Codescalers: Software Engineer** (2016 – 2020) Designed and built cloud tooling for VM lifecycle management, container orchestration, and infrastructure automation.
- **Otlob.com: Data Analyst** (Freelance) Built audio fingerprinting pipeline over large-scale video datasets to measure marketing campaign reach.
- **Orascom Telecom: Web Developer** (Freelance) Built an internal PHP/CodeIgniter application for server room environmental monitoring, replacing manual Excel workflows.

Key Projects

Operating Systems & Core Infrastructure

- **Zero-OS:** Leading development of the custom Linux-based OS powering the entire Threefold Grid (Go)
- **ContentHive:** Content-addressed storage system inspired by IPFS (Go)
- **0-Disk:** Network Block Device server with pluggable storage backends (Tarantool, Lua)
- **OpenVCloud:** Monitoring, alerting, and hardware health-check tooling for a libvirt-based cloud platform (Python)

Platforms & Developer Tools

- **KubeCloud:** Highly available Kubernetes platform on the Threefold Grid (Go, Vue.js, PostgreSQL)
- **VDC:** Virtual Data Center — managed Kubernetes clusters, S3 storage, Prometheus/Grafana monitoring (Python, Stellar, Ansible)
- **VDC Marketplace:** One-click application deployment marketplace on top of VDC
- **AIBroker:** AI gateway aggregating multiple LLM providers with MCP server integration (Rust, Go)
- **Terraform Provider:** Infrastructure-as-Code plugin for provisioning on the Threefold Grid (Go)
- **TFRobot:** Mass-deployment tool for large VM fleets (Go)
- **Gridify:** Heroku-like deploy tool for the Threefold Grid (Go)
- **Farmerbot:** Power-management and operations automation for grid farmers (Go)

SDKs & Frameworks

- **Go SDK:** Primary SDK enabling delivery of multiple grid-based projects (Go)
- **TypeScript SDK:** Web developer SDK for Threefold Grid integration (TypeScript)
- **JS-NG / JS-SDK:** Cloud automation framework and Threebot application server (Python, Gevent)
- **EWf:** Embeddable workflow framework for resilient, stateful multi-step processes (Go)
- **CRM:** CRM system for Threefold Token Foundation (Python, Flask, GraphQL, PostgreSQL)
- **publishingtools:** Web publishing framework for wikis, blogs, and websites (Crystal, Kemal, Svelte)

Infrastructure Tooling

- **AYS:** Agentless Infrastructure-as-Code orchestration — event systems, background tasks, and service templates
- **Go farmer:** Cross-platform GUI for farm management (Go, Fyne)

- Additional: OAuth provider for Gogs, DMIDecode parser, GitHub–Gogs sync, documentation generators for Jumpscale

Open Source Projects

- tcprouter: TCP router with SNI-based routing (Go)
- tfgateway: Threefold gateway built on tcprouter and CoreDNS Redis plugin
- coredns-redis: Redis plugin for CoreDNS
- raqeeb: PID 1 replacement and advanced alternative to Foreman
- pygundb: GUN.js database server and client (Python)
- pyparsec: Parser combinator library for Python
- python-sonic-client: Python client for Sonic search backend
- python-pcre: PCRE bindings for Python using ctypes
- flask_itsyouonline: OAuth2 middleware for Flask
- nim-servy: Web framework for Nim
- nim-redisclient: Redis client and parser for Nim
- nim-assets: Bundle assets into a single Nim binary
- vscode-capnp: Cap'n Proto extension for VS Code
- helloED: Pluggable, fast PyGTK text editor
- stellargui: Lightweight Stellar blockchain wallet
- Pida IDE: Contributed RegexToolkit and Pastebin plugins
- raml2spore: Generate SPORE clients from RAML specifications

Publications / Translations

- Nim Days: Practical applications and libraries in Nim
- Python: Extensive Guide
- Introduction to Ruby
- Translations: PyQt4, PyGTK, Mono IronPython, and Cairo tutorials translated into Arabic

Skills & Technologies

- **Languages:** Go, Python, Rust, TypeScript, C, Nim, Crystal, Bash
- **Databases:** PostgreSQL, Redis, MySQL, MongoDB, InfluxDB
- **Infrastructure:** Linux, Kubernetes, K3s, Helm, Terraform/OpenTofu, Ansible, Docker, Prometheus, Grafana, CoreDNS, Nginx, Traefik
- **APIs & Protocols:** REST, GraphQL, gRPC, Cap'n Proto
- **Frontend:** Vue, Svelte, React

Education

- BSc in Electrical Engineering, Helwan University, Cairo, Egypt
- English: Business proficiency